Floorball Match Predictor

Machine learning project for the subject of PV

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1. Introduction

This documentation describes the development of the Floorball Match Predictor as a software project designed to predict the outcome of floorball matches using machine learning. The system collects real-world data from multiple floorball leagues, processes it into structured datasets, trains predictive models, and provides a graphical interface for predicting match results.

The project consists of:

- Data collection (API + web scraping)
- Data preprocessing and feature engineering
- Machine learning model training and evaluation
- Desktop UI application for predictions

2. User specifications and requirements

The system is designed to provide the following functionality:

- Select a floorball competition (league)
- Select home and away teams
- Predict match outcome (probability of each team winning)

Technical requirements:

- Python runtime environment
- Required libraries installed
- Pre-generated datasets and trained model

3. Application architecture

The application is divided into four main components:

Data Collection Layer

Responsible for acquiring real-world data:

- API communication (Sportradar)
- Web scraping (multiple league websites)

Data Processing Layer

Transforms raw data into structured datasets:

- Cleaning and filtering data
- Feature engineering
- Dataset creation for ML

Machine Learning Layer

Handles training and prediction:

- Model training
- Evaluation
- Model persistence

Application Layer (UI)

User interface built with customtkinter, allowing users to interact with the system and obtain predictions.

4. Behavioral logic

Data collection process

The data collection process is divided into two main parts. Match data collection and player statistics collection.

Match data collection (API)

1. The system sends requests to the Sportradar API to retrieve available floorball competitions.

2. For each selected competition, all available seasons are downloaded.

3. For each season, match summaries are requested in batches.

4. Only finished matches are processed, ensuring valid target labels.

5. For each match, the following attributes are extracted:

    - Teams (home and away)

    - Scores

    - Match timestamp

    - Competition and season information

6. The collected data is stored in a raw dataset for further processing.

Player statistics collection (web scraping)

In addition to match data, the system collects detailed player statistics from official league websites using web scraping.

1. A browser automation tool (Playwright) is used to simulate user interaction.

2. The system navigates to official league statistics pages.
3. Filters are applied (season, competition, teams).
4. Tables containing player statistics are located and parsed.
5. For each player, the following attributes are extracted:
    - Player name
    - Team name
    - Matches played
    - Goals
    - Assists
    - Points
    - Additional statistics (shots, penalties, etc.)
6. Data is collected across multiple leagues:
    - Czech Extraliga
    - Finnish F-liiga
    - Swedish SSL
7. The collected player data is saved into CSV files.

Data integration

After both datasets are collected:

1. Player statistics are aggregated on a team level.
2. A new feature called roster strength is computed based on player performance.
3. This feature is merged with the match dataset.
4. The final dataset is used for machine learning training.

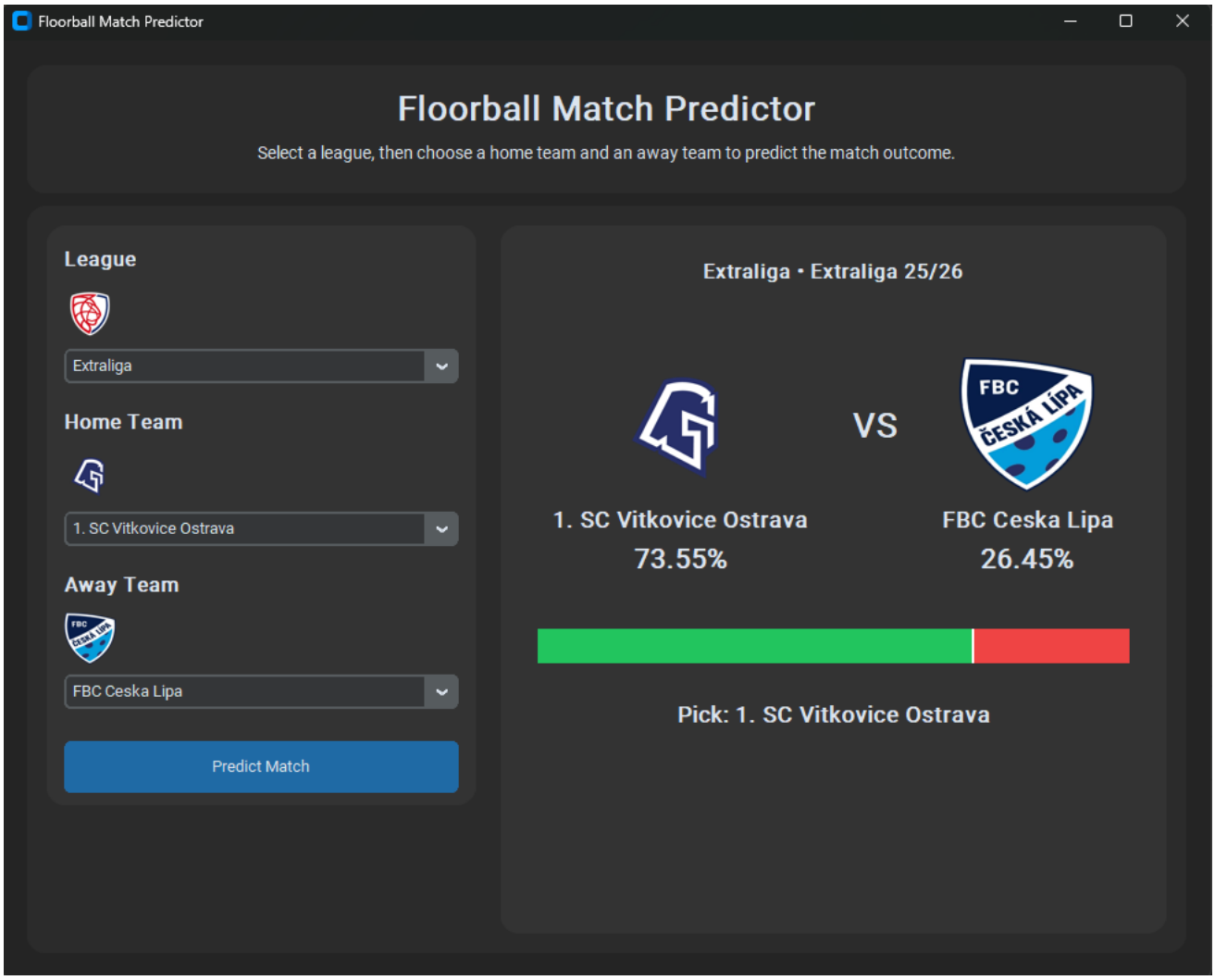
Data processing process

1. Matches are sorted chronologically
2. For each match, historical statistics are calculated:
    a. Team points

   b. Goals scored and conceded
   c. Recent performace (last 5 matches)
3. Rankings are dynamically calculated
4. Feature vectors are generated for machine learning

Prediction process

1. User selects teams.
2. System computes current statistics.
3. Feature vector is created.
4. Model predicts probability.
5. Result is displayed in UI.



Picture: UI of the application

5. Dataset and data collection

API (primary source)

- Sportradar Floorball API
- Used to obtain:
    - Match results
    - Competition structure
    - Seasons and teams

Web scraping (secondary source)

Data is scraped using Playwright, simulating a real browser.

Used to obtain player statistics:

- Czech Extraliga
- Finnish F-Liiga
- Swedish SSL

6. Data preprocessing and feature engineering

Raw data cannot be used directly for machine learning and must be processed.

Cleaning

- Removal of incomplete matches
- Conversion of data types (strings to numeric)
- Handling missing values

Transformation

- Standardization of season formats
- Normalization of team names

Feature engineering

The following features are computed:

- Team ranking
- Points difference
- Goal difference
- Recent form (last 5 matches)
- Home vs away performance
- Average goals scored and conceded

7. Machine Learning model

Problem definition

Binary classification:

- Target: whether the home team wins

Models used

- Logistic regression
- Random forest

Pipeline structure

- Missing values – SimpleImputer
- Scaling – StandardScaler
- Encoding – OneHotEncoder

Training process

1. Dataset is split into training and testing sets (80/20)
2. Models are trained on historical data

8. Error handling

The system includes multiple error-handling mechanisms:

- API rate limit handling (retry logic)
- File existence check
- Safe processing of missing values

9. Third party libraries

The project maintains a structured and modular design by utilizing several third-party libraries and frameworks. These libraries are used strictly for well-defined purposes such as data processing, machine learning, communication, and user interface development.

- pandas: Used for efficient data manipulation and analysis. The library enables loading, transforming, filtering, and aggregating large datasets. It plays a key role in both preprocessing and feature engineering stages of the project.
- scikit-learn: Provides machine learning functionality including model training, evaluation, and preprocessing. The project uses:
    - Logistic Regression and Random Forest models
    - Data preprocessing tools such as StandardScaler, OneHotEncoder, and SimpleImputer
    - Evaluation metrics including accuracy, ROC AUC, confusion matrix, and classification report
- joblib: Used for saving and loading trained machine learning models. This allows the trained model to be reused in the application without retraining.
- requests: Handles HTTP communication with the Sportradar API. It is used to download match data, including competitions, seasons, and match summaries.
- playwright: Used for web scraping dynamic websites. It simulates a real browser environment and allows interaction with web pages, including clicking elements, selecting filters, and extracting structured data from tables.
- customtkinter: Provides a modern graphical user interface framework built on top of Tkinter. It is used to create the desktop application interface, including:
    - Dropdown selection of teams and competitions
    - Display of predictions and probabilities
    - Visual elements such as graphs and logos
- python-dotenv: Used to securely load environment variables (API key) from a .env file, preventing sensitive data from being hardcoded.
- Pillow (PIL): Used for loading and displaying images (team logos and competition logos) within the graphical interface.

10. Project summary

The **Floorball Match Predictor** project demonstrates a complete software development lifecycle, from raw data acquisition to final user interaction.

- Real-world data collection (API and web scraping)
- Data preprocessing and feature engineering
- Machine learning model training and evaluation
- Match outcome prediction system
- Desktop application with graphical user interface

Developed as a school project for PV subject. Serves as a foundation for further improvements and extensions.

Střední průmyslová škola elektrotechnická website: https://www.spsejecna.cz/
Github repository: https://github.com/Forkxel/Floorball-Match-Predictor

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